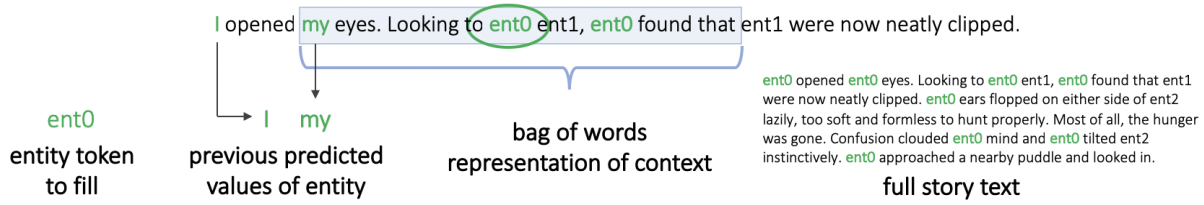

Figure 3: Input for Coreferent entity reference generation. The model has a representation of the entity context in a bag of words form, all previous predicted values for the same anonymized entity token, and the full text story. The green circle represents the entity mention the model is attempting to fill.

To train the pointer mechanism, the final hidden state of the model h is used as input to a classifier $p_{\text{copy}}(h) = \sigma(w_{\text{copy}} \cdot h)$. w_{copy} is a fixed dimension parameter vector. When the model classifier predicts to copy, the previously decoded abstract entity token with the maximum attention value is copied. One head of the decoder multi-head self-attention mechanism is used as the pointer copy attention head, to allow the heads to specialize.

Sub-word Modelling Entities are often rare or novel words, so word-based vocabularies can be inadequate. We compare entity generation using word-based, byte-pair encoding (BPE) (Sennrich et al., 2015), and character-level models.

NER-based Entity Reference Generation Here, each placeholder string should map onto one (possibly multiword) surface form—e.g. all occurrences of the placeholder *ent0* should map only a single string, such as *Bilbo Baggins*. We train a simple model that maps a combination placeholder token and story (with anonymized entities) to the surface form of the placeholder. While the placeholder can appear multiple times, we only make one prediction for each placeholder as they all correspond to the same string.

Coreference-based Entity Reference Generation Generating entities based on coreference clusters is more challenging than for our NER entity clusters, because different mentions of the same entity may use different surface forms. We generate a separate reference for each mention by adding the following inputs to the above model:

- A *bag-of-words* context window around the specific entity mention, which allows local context to determine if an entity should be a name, pronoun or nominal reference.
- *Previously generated* references for the same entity placeholder. For example, if the model is filling in the third instance of *ent0*, it receives that the previous two generations for *ent0* were *Bilbo, him*. Providing the previous entities allows the model to maintain greater consistency between generations.

5 Experimental Setup

5.1 Data

We use the WRITINGPROMPTS dataset from (Fan et al., 2018) [4] of 300k story premises paired with long stories. Stories are on average 734 words, making the generation far longer compared to related work on storyline generation. In this work, we focus on the prompt to story generation aspect of this task. We assume models receive a human-written prompt, as shown in Figure 1. We follow the previous preprocessing of limiting stories to 1000 words and fixing the vocabulary size to 19,025 for prompts and 104,960 for stories.

5.2 Baselines

We compare our results to the *Fusion* model from Fan et al. (2018) which generates the full story directly from the prompt. We also implement various decomposition strategies as baselines:

- *Summarization*: We propose a new baseline that generates a summary conditioned upon the prompt and then a story conditioned upon the summary. Story summaries are obtained with a multi-sentence summarization model (Wu et al., 2019) trained on the full-text version of the CNN-Dailymail summarization corpus (Hermann et al., 2015; Nallapati et al., 2016; See et al., 2017)[5] and applied to stories.
- *Keyword Extraction*: We generate a series of keywords conditioned upon the prompt and

[4] https://github.com/pytorch/fairseq/tree/master/examples/stories

[5] https://github.com/abisee/cnn-dailymail

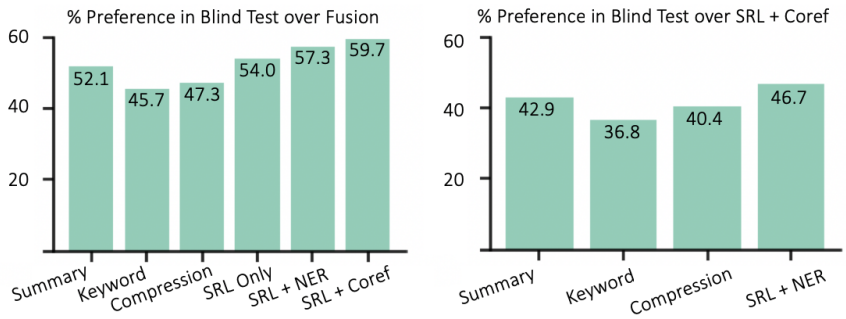

Figure 4: Human evaluations of different decomposed models for story generation. We find that using SRL action plans and coreference-resolution to build entity clusters generates stories that are preferred by human judges.

Decomposition	**Stage 1** $-\log p(z^*)$	**Stage 2** $-\log p(x\|z^*)$
Summary	4.20	5.09
Keyword	6.92	4.23
Compression	5.05	3.64
SRL Action Plan	2.72	3.95
NER Entity Anonymization	3.32	4.75
Coreference Anonymization	3.15	4.55

Table 1: Negative log likelihood of generating stories using different decompositions (lower is easier for the model). Stage 1 is the generation of the intermediate representation z^*, and Stage 2 is the generation of the story x conditioned upon z^*. Entity generation is with a word-based vocabulary to be consistent with the other models.

then a story conditioned upon the keywords, based on Yao et al. (2019). Following Yao et al, we extract keywords with the RAKE algorithm (Rose et al., 2010)[6]. Yao et al. extract one word per sentence, but we find that extracting $n = 10$ keyword phrases per story worked well, as our stories are much longer.

- *Sentence Compression*: Inspired by Xu et al. (2018), we generate a story with compressed sentences conditioned upon the prompt and then a story conditioned upon the compression. We use the same deletion-based compression data as Xu et al., from Filippova and Altun (2013)[7]. We train a seq2seq model to compress all non-dialog story sentences (as the training data does not contain much spoken dialogue). The compressed sentences are concatenated to form the compressed story.

[6] https://pypi.org/project/rake-nltk/

[7] https://github.com/google-research-datasets/sentence-compression

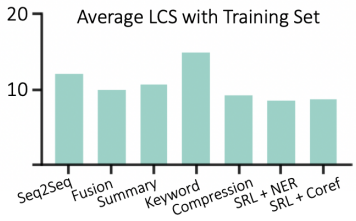

Figure 5: Average Longest Common Subsequence of Generated Stories with human-written stories in the training set.

5.3 Training

We implement models using `fairseq-py` (Ott et al., 2019)[8] in PyTorch and train Fan et al. (2018)'s convolutional architecture. We tune all hyperparameters on validation data.

5.4 Generation

We suppress the generation of unknown tokens to ease human evaluation. For all evaluations, we require stories to be at least 150 words and cut off the story at the nearest sentence for stories longer than 250 words. We generate stories with temperature 0.8 and random top-k sampling method proposed in (Fan et al., 2018), where next words are sampled from the top k candidates rather than the entire vocabulary distribution. We set $k = 10$.

6 Experiments

6.1 Comparing Decomposition Strategies

Automated Evaluation We compare the relative difficulty of modelling using each decomposition strategy by measuring the log loss of the different stages in Table 1. We observe that generating the SRL structure has a lower negative log-likelihood and so is much easier than generating

[8] https://github.com/pytorch/fairseq/

Fusion Baseline Generated Story

I was born with my own name, a name the world has never seen before. I'm a scientist, and I am a scientist, and I am the greatest inventor ever conceived.
It is said that in the time I was a child, that the world had been born.
It is said that the world has been in peril for nearly a year and the city has grown.
There's a lot we can do about the new world, and that will cause the new world to become the only thing with us.
It has been a year since the world began to turn around, and people were afraid of this new world.
It took us a year to find this world again.

SRL + Coref Generated Story

It was a dark and stormy night.
There was no one here, and the sky above the stars was as dark and devoid of all but the dark, dark and empty sky.
It was just a dark, stormy night, and there was the quiet, and silent sound of the wind. ``Where are you?" I called for him.
> There was a pause, and a long silence. There was nothing. And there was nothing, except for a light, a small circle in the center of the circle.
`` Who are you, and where are you now? '' I said. In the center was a small wooden circle. A man with a black shirt and a circle was there, and one man with two wings.
He looked as if he was about to die. Inside the circle was the silhouette of the wing.

Figure 6: Our decomposition can generate more coherent stories than previous work.

either summaries, keywords, or compressed sentences — a benefit of its more structured form. We find keyword generation is especially difficult as the identified keywords are often the more salient, rare words appearing in the story, which are challenging for neural seq2seq models to generate. This result suggests that rare words should appear mostly at the last levels of the decomposition. Further, we compare models with entity-anonymized stories as an intermediate representation, either with NER-based or coreference-based entity anonymization. Entity references are then filled using a word-based model.[9] Perhaps surprisingly, naming entities proves more difficult than creating the entity-anonymized stories—providing insight into the relative difficulty of different subproblems of story generation.

Finally, we analyze the similarity of the generated stories with the stories in the training set. We quantify this by measuring the maximum and average longest common subsequence of tokens of a generated story with all human-written stories from the training set. High LCS values would indicate models are copying large subparts from existing stories rather than creatively writing new stories. Results shown in Figure 5 indicate that our proposed decomposition copies slightly less long sequences from the training set compared to the baselines — by separating verb and entity generation into distinct parts, we generate fewer long sequences already present in the training set.

[9]To make likelihoods are comparable across models.

Human Evaluation To compare overall story quality using various decomposition strategies, we conduct human evaluation using a crowdworking platform. Judges are shown two different stories that were generated based on the same human-written prompt (but do not see the prompt). Evaluators are asked to mark which story they prefer. 100 stories are evaluated for each model by 3 different judges. To reduce variance, stories from all models are trimmed to 200 words.

Figure 6 shows that human evaluators prefer our novel decompositions over a carefully tuned Fusion model from Fan et al. (2018) by about 60% in a blind comparison. We see additive gains from modelling actions and entities. In a second study, evaluators compared various baselines against stories generated by our strongest model, which uses SRL-based action plans and coreference-based entity anonymization. In all cases, our full decomposition is preferred.

6.2 Effect of SRL Decomposition

Human-written stories feature a wide variety of events, while neural models are plagued by generic generations and repetition. Table 2 quantifies model performance on two metrics to assess action diversity: (1) the number of unique verbs generated, averaged across all stories (2) the percentage of diverse verbs, measured by the percent of all verbs generated in the test set that are not one of the top 5 most frequent verbs. A higher percentage indicates more diverse events.[10]

Our decomposition using the SRL predicate-argument structure improves the model's ability to generate diverse verbs. Adding verb attention leads to further improvement. Qualitatively, the model can often outline clear action sequences, as shown in Figure 7. However, all models remain far from matching the diversity of human stories.

6.3 Comparing Entity Reference Models

We explored a variety of different ways to generate the full text of abstracted entities—using different amounts of context and different granularities of subword generation. To compare these models, we calculated their accuracy at predicting the correct reference in Table 3. Each model evaluates $n = 10, 50, 100$ different entities in the test set, 1 real and $n-1$ randomly sampled distractors. Mod-

[10]We identify verbs using Spacy: https://spacy.io/